0 as r goes to infinity. It is important that the bound $n/(n+1)$ does not depend on the shape of the simplex since repeated barycentric subdivision produces simplices of many different shapes.

(2) *Barycentric Subdivision of Linear Chains.* The main part of the proof will be to construct a subdivision operator $S: C_n(X) \to C_n(X)$ and show this is chain homotopic to the identity map. First we will construct S and the chain homotopy in a more restricted linear setting.

For a convex set Y in some Euclidean space, the linear maps $\Delta^n \to Y$ generate a subgroup of $C_n(Y)$ that we denote $LC_n(Y)$, the *linear chains*. The boundary map $\partial: C_n(Y) \to C_{n-1}(Y)$ takes $LC_n(Y)$ to $LC_{n-1}(Y)$, so the linear chains form a subcomplex of the singular chain complex of Y. We can uniquely designate a linear map $\lambda: \Delta^n \to Y$ by $[w_0, \cdots, w_n]$ where w_i is the image under λ of the i^{th} vertex of Δ^n. To avoid having to make exceptions for 0-simplices it will be convenient to augment the complex $LC(Y)$ by setting $LC_{-1}(Y) = \mathbb{Z}$ generated by the empty simplex $[\varnothing]$, with $\partial[w_0] = [\varnothing]$ for all 0-simplices $[w_0]$.

Each point $b \in Y$ determines a homomorphism $b: LC_n(Y) \to LC_{n+1}(Y)$ defined on basis elements by $b([w_0, \cdots, w_n]) = [b, w_0, \cdots, w_n]$. Geometrically, the homomorphism b can be regarded as a cone operator, sending a linear chain to the cone having the linear chain as the base of the cone and the point b as the tip of the cone. Applying the usual formula for ∂, we obtain the relation $\partial b([w_0, \cdots, w_n]) = [w_0, \cdots, w_n] - b(\partial[w_0, \cdots, w_n])$. By linearity it follows that $\partial b(\alpha) = \alpha - b(\partial\alpha)$ for all $\alpha \in LC_n(Y)$. This expresses algebraically the geometric fact that the boundary of a cone consists of its base together with the cone on the boundary of its base. The relation $\partial b(\alpha) = \alpha - b(\partial\alpha)$ can be rewritten as $\partial b + b\partial = \mathbb{1}$, so b is a chain homotopy between the identity map and the zero map on the augmented chain complex $LC(Y)$.

Now we define a subdivision homomorphism $S: LC_n(Y) \to LC_n(Y)$ by induction on n. Let $\lambda: \Delta^n \to Y$ be a generator of $LC_n(Y)$ and let b_λ be the image of the barycenter of Δ^n under λ. Then the inductive formula for S is $S(\lambda) = b_\lambda(S\partial\lambda)$ where $b_\lambda: LC_{n-1}(Y) \to LC_n(Y)$ is the cone operator defined in the preceding paragraph. The induction starts with $S([\varnothing]) = [\varnothing]$, so S is the identity on $LC_{-1}(Y)$. It is also the identity on $LC_0(Y)$, since when $n = 0$ the formula for S becomes $S([w_0]) = w_0(S\partial[w_0]) = w_0(S([\varnothing])) = w_0([\varnothing]) = [w_0]$. When λ is an embedding, with image a genuine n-simplex $[w_0, \cdots, w_n]$, then $S(\lambda)$ is the sum of the n-simplices in the barycentric subdivision of $[w_0, \cdots, w_n]$, with certain signs that could be computed explicitly. This is apparent by comparing the inductive definition of S with the inductive definition of the barycentric subdivision of a simplex.

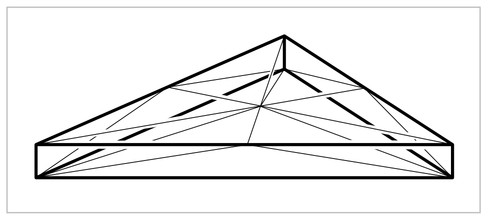
Let us check that the maps S satisfy $\partial S = S\partial$, and hence give a chain map from the chain complex $LC(Y)$ to itself. Since $S = \mathbb{1}$ on $LC_0(Y)$ and $LC_{-1}(Y)$, we certainly have $\partial S = S\partial$ on $LC_0(Y)$. The result for larger n is given by the following calculation, in which we omit some parentheses to unclutter the formulas:

$$
\begin{aligned}
\partial S\lambda &= \partial(b_\lambda(S\partial\lambda)) \\
&= S\partial\lambda - b_\lambda(\partial S\partial\lambda) \qquad \text{since } \partial b_\lambda + b_\lambda\partial = \mathbb{1} \\
&= S\partial\lambda - b_\lambda(S\partial\partial\lambda) \qquad \text{by induction on } n \\
&= S\partial\lambda \qquad \text{since } \partial\partial = 0
\end{aligned}
$$

We next build a chain homotopy $T: LC_n(Y) \to LC_{n+1}(Y)$ between S and the identity, fitting into a diagram

$$
\begin{tikzcd}
\cdots \arrow[r] & LC_2(Y) \arrow[r] \arrow[d, "S"] & LC_1(Y) \arrow[r] \arrow[d, "S"] \arrow[dl, "T"'] & LC_0(Y) \arrow[r] \arrow[d, "S"', "\mathbb{1}"] \arrow[dl, "T"'] & LC_{-1}(Y) \arrow[r] \arrow[d, "S"', "\mathbb{1}"] \arrow[dl, "T"', "0"] & 0 \\
\cdots \arrow[r] & LC_2(Y) \arrow[r] & LC_1(Y) \arrow[r] & LC_0(Y) \arrow[r] & LC_{-1}(Y) \arrow[r] & 0
\end{tikzcd}
$$

We define T on $LC_n(Y)$ inductively by setting $T = 0$ for $n = -1$ and letting $T\lambda = b_\lambda(\lambda - T\partial\lambda)$ for $n \geq 0$. The geometric motivation for this formula is an inductively defined subdivision of $\Delta^n \times I$ obtained by joining all simplices in $\Delta^n \times \{0\} \cup \partial\Delta^n \times I$ to the barycenter of $\Delta^n \times \{1\}$, as indicated in the figure in the case $n = 2$. What T actually does is take the image of this subdivision under the projection $\Delta^n \times I \to \Delta^n$.



The chain homotopy formula $\partial T + T\partial = \mathbb{1} - S$ is trivial on $LC_{-1}(Y)$ where $T = 0$ and $S = \mathbb{1}$. Verifying the formula on $LC_n(Y)$ with $n \geq 0$ is done by the calculation

$$
\begin{aligned}
\partial T\lambda &= \partial(b_\lambda(\lambda - T\partial\lambda)) \\
&= \lambda - T\partial\lambda - b_\lambda(\partial(\lambda - T\partial\lambda)) \qquad \text{since } \partial b_\lambda = \mathbb{1} - b_\lambda\partial \\
&= \lambda - T\partial\lambda - b_\lambda(S\partial\lambda + T\partial\partial\lambda) \qquad \text{by induction on } n \\
&= \lambda - T\partial\lambda - S\lambda \qquad \text{since } \partial\partial = 0 \text{ and } S\lambda = b_\lambda(S\partial\lambda)
\end{aligned}
$$

Now we are done with inductive arguments and we can discard the group $LC_{-1}(Y)$ which was used only as a convenience. The relation $\partial T + T\partial = \mathbb{1} - S$ still holds without $LC_{-1}(Y)$ since T was zero on $LC_{-1}(Y)$.

(3) *Barycentric Subdivision of General Chains.* Define $S: C_n(X) \to C_n(X)$ by setting $S\sigma = \sigma_\sharp S\Delta^n$ for a singular n-simplex $\sigma: \Delta^n \to X$. Since $S\Delta^n$ is the sum of the n-simplices in the barycentric subdivision of Δ^n, with certain signs, $S\sigma$ is the corresponding signed sum of the restrictions of σ to the n-simplices of the barycentric subdivision of Δ^n. The operator S is a chain map since

$$
\begin{aligned}
\partial S\sigma &= \partial\sigma_\sharp S\Delta^n = \sigma_\sharp \partial S\Delta^n = \sigma_\sharp S\partial\Delta^n \\
&= \sigma_\sharp S\big(\textstyle\sum_i (-1)^i \Delta_i^n\big) \qquad \text{where } \Delta_i^n \text{ is the } i^{th} \text{ face of } \Delta^n \\
&= \textstyle\sum_i (-1)^i \sigma_\sharp S\Delta_i^n \\
&= \textstyle\sum_i (-1)^i S(\sigma|\Delta_i^n) \\
&= S\big(\textstyle\sum_i (-1)^i \sigma|\Delta_i^n\big) = S(\partial\sigma)
\end{aligned}
$$

In similar fashion we define $T:C_n(X)\to C_{n+1}(X)$ by $T\sigma = \sigma_\sharp T\Delta^n$, and this gives a chain homotopy between S and the identity, since the formula $\partial T + T\partial = \mathbb{1} - S$ holds by the calculation

$$\begin{aligned}\partial T\sigma = \partial\sigma_\sharp T\Delta^n = \sigma_\sharp \partial T\Delta^n = \sigma_\sharp(\Delta^n - S\Delta^n - T\partial\Delta^n) &= \sigma - S\sigma - \sigma_\sharp T\partial\Delta^n \\ &= \sigma - S\sigma - T(\partial\sigma)\end{aligned}$$

where the last equality follows just as in the previous displayed calculation, with S replaced by T.

(4) *Iterated Barycentric Subdivision.* A chain homotopy between $\mathbb{1}$ and the iterate S^m is given by the operator $D_m = \sum_{0\le i<m} TS^i$ since

$$\begin{gathered}\partial D_m + D_m\partial = \sum_{0\le i<m}(\partial TS^i + TS^i\partial) = \sum_{0\le i<m}(\partial TS^i + T\partial S^i) = \\ \sum_{0\le i<m}(\partial T + T\partial)S^i = \sum_{0\le i<m}(\mathbb{1} - S)S^i = \sum_{0\le i<m}(S^i - S^{i+1}) = \mathbb{1} - S^m\end{gathered}$$

For each singular n-simplex $\sigma:\Delta^n\to X$ there exists an m such that $S^m(\sigma)$ lies in $C_n^{\mathcal{U}}(X)$ since the diameter of the simplices of $S^m(\Delta^n)$ will be less than a Lebesgue number of the cover of Δ^n by the open sets $\sigma^{-1}(\operatorname{int} U_j)$ if m is large enough. (Recall that a Lebesgue number for an open cover of a compact metric space is a number $\varepsilon > 0$ such that every set of diameter less than ε lies in some set of the cover; such a number exists by an elementary compactness argument.) We cannot expect the same number m to work for all σ's, so let us define $m(\sigma)$ to be the smallest m such that $S^m\sigma$ is in $C_n^{\mathcal{U}}(X)$.

Suppose we define $D:C_n(X)\to C_{n+1}(X)$ by $D\sigma = D_{m(\sigma)}\sigma$. To see whether D is a chain homotopy, we manipulate the chain homotopy equation

$$\partial D_{m(\sigma)}\sigma + D_{m(\sigma)}\partial\sigma = \sigma - S^{m(\sigma)}\sigma$$

into an equation whose left side is $\partial D\sigma + D\partial\sigma$ by moving the second term on the left side to the other side of the equation and adding $D\partial\sigma$ to both sides:

$$\partial D\sigma + D\partial\sigma = \sigma - \big[S^{m(\sigma)}\sigma + D_{m(\sigma)}(\partial\sigma) - D(\partial\sigma)\big]$$

If we define $\rho(\sigma)$ to be the expression in brackets in this last equation, then this equation has the form

$$(*) \qquad \partial D\sigma + D\partial\sigma = \sigma - \rho(\sigma)$$

We claim that $\rho(\sigma)\in C_n^{\mathcal{U}}(X)$. This is obvious for the term $S^{m(\sigma)}\sigma$. For the remaining part $D_{m(\sigma)}(\partial\sigma) - D(\partial\sigma)$, note first that if σ_j denotes the restriction of σ to the j^{th} face of Δ^n, then $m(\sigma_j)\le m(\sigma)$, so every term $TS^i(\sigma_j)$ in $D(\partial\sigma)$ will be a term in $D_{m(\sigma)}(\partial\sigma)$. Thus $D_{m(\sigma)}(\partial\sigma) - D(\partial\sigma)$ is a sum of terms $TS^i(\sigma_j)$ with $i\ge m(\sigma_j)$, and these terms lie in $C_n^{\mathcal{U}}(X)$ since T takes $C_{n-1}^{\mathcal{U}}(X)$ to $C_n^{\mathcal{U}}(X)$

We can thus regard the equation $(*)$ as defining $\rho:C_n(X)\to C_n^{\mathcal{U}}(X)$. For varying n these ρ's form a chain map since $(*)$ implies $\partial\rho(\sigma) = \partial\sigma - \partial D\partial(\sigma) = \rho(\partial\sigma)$.

The equation $(*)$ says that $\partial D + D\partial = \mathbb{1} - \iota\rho$ for $\iota : C_n^{\mathcal{U}}(X) \hookrightarrow C_n(X)$ the inclusion. Furthermore, $\rho\iota = \mathbb{1}$ since D is identically zero on $C_n^{\mathcal{U}}(X)$, as $m(\sigma) = 0$ if σ is in $C_n^{\mathcal{U}}(X)$, hence the summation defining $D\sigma$ is empty. Thus we have shown that ρ is a chain homotopy inverse for ι. $\square$

Proof of the Excision Theorem: We prove the second version, involving a decomposition $X = A \cup B$. For the cover $\mathcal{U} = \{A, B\}$ we introduce the suggestive notation $C_n(A+B)$ for $C_n^{\mathcal{U}}(X)$, the sums of chains in A and chains in B. At the end of the preceding proof we had formulas $\partial D + D\partial = \mathbb{1} - \iota\rho$ and $\rho\iota = \mathbb{1}$. All the maps appearing in these formulas take chains in A to chains in A, so they induce quotient maps when we factor out chains in A. These quotient maps automatically satisfy the same two formulas, so the inclusion $C_n(A+B)/C_n(A) \hookrightarrow C_n(X)/C_n(A)$ induces an isomorphism on homology. The map $C_n(B)/C_n(A\cap B) \to C_n(A+B)/C_n(A)$ induced by inclusion is obviously an isomorphism since both quotient groups are free with basis the singular n-simplices in B that do not lie in A. Hence we obtain the desired isomorphism $H_n(B, A\cap B) \approx H_n(X, A)$ induced by inclusion. $\square$

All that remains in the proof of Theorem 2.13 is to replace relative homology groups with absolute homology groups. This is achieved by the following result.

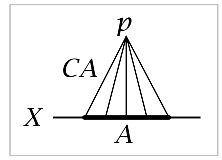
Proposition 2.22. *For good pairs (X, A), the quotient map $q : (X, A) \to (X/A, A/A)$ induces isomorphisms $q_* : H_n(X, A) \to H_n(X/A, A/A) \approx \widetilde{H}_n(X/A)$ for all n.*

Proof: Let V be a neighborhood of A in X that deformation retracts onto A. We have a commutative diagram

$$\begin{tikzcd}
H_n(X,A) \arrow[r] \arrow[d, "q_*"] & H_n(X,V) \arrow[d, "q_*"] & H_n(X-A, V-A) \arrow[l] \arrow[d, "q_*"] \\
H_n(X/A, A/A) \arrow[r] & H_n(X/A, V/A) & H_n(X/A-A/A, V/A-A/A) \arrow[l]
\end{tikzcd}$$

The upper left horizontal map is an isomorphism since in the long exact sequence of the triple (X, V, A) the groups $H_n(V, A)$ are zero for all n, because a deformation retraction of V onto A gives a homotopy equivalence of pairs $(V, A) \simeq (A, A)$, and $H_n(A, A) = 0$. The deformation retraction of V onto A induces a deformation retraction of V/A onto A/A, so the same argument shows that the lower left horizontal map is an isomorphism as well. The other two horizontal maps are isomorphisms directly from excision. The right-hand vertical map q_* is an isomorphism since q restricts to a homeomorphism on the complement of A. From the commutativity of the diagram it follows that the left-hand q_* is an isomorphism. $\square$

This proposition shows that relative homology can be expressed as reduced absolute homology in the case of good pairs (X, A), but in fact there is a way of doing this for arbitrary pairs. Consider the space $X \cup CA$ where CA is the cone $(A \times I)/(A \times \{0\})$



whose base $A\times\{1\}$ we identify with $A \subset X$. Using terminology introduced in Chapter 0, $X\cup CA$ can also be described as the mapping cone of the inclusion $A \hookrightarrow X$. The assertion is that $H_n(X,A)$ is isomorphic to $\widetilde{H}_n(X \cup CA)$ for all n via the sequence of isomorphisms

$$\widetilde{H}_n(X \cup CA) \approx H_n(X \cup CA, CA) \approx H_n(X \cup CA - \{p\}, CA - \{p\}) \approx H_n(X, A)$$

where $p \in CA$ is the tip of the cone. The first isomorphism comes from the exact sequence of the pair, using the fact that CA is contractible. The second isomorphism is excision, and the third isomorphism comes from the deformation retraction of $CA - \{p\}$ onto A.

Here is an application of the preceding proposition:

Example 2.23. Let us find explicit cycles representing generators of the infinite cyclic groups $H_n(D^n, \partial D^n)$ and $\widetilde{H}_n(S^n)$. Replacing $(D^n, \partial D^n)$ by the equivalent pair $(\Delta^n, \partial\Delta^n)$, we will show by induction on n that the identity map $i_n : \Delta^n \to \Delta^n$, viewed as a singular n-simplex, is a cycle generating $H_n(\Delta^n, \partial\Delta^n)$. That it is a cycle is clear since we are considering relative homology. When $n = 0$ it certainly represents a generator. For the induction step, let $\Lambda \subset \Delta^n$ be the union of all but one of the $(n-1)$-dimensional faces of Δ^n. Then we claim there are isomorphisms

$$H_n(\Delta^n, \partial\Delta^n) \xrightarrow{\approx} H_{n-1}(\partial\Delta^n, \Lambda) \xleftarrow{\approx} H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1})$$

The first isomorphism is a boundary map in the long exact sequence of the triple $(\Delta^n, \partial\Delta^n, \Lambda)$, whose third terms $H_i(\Delta^n, \Lambda)$ are zero since Δ^n deformation retracts onto Λ, hence $(\Delta^n, \Lambda) \simeq (\Lambda, \Lambda)$. The second isomorphism comes from the preceding proposition since we are dealing with good pairs and the inclusion $\Delta^{n-1} \hookrightarrow \partial\Delta^n$ as the face not contained in Λ induces a homeomorphism of quotients $\Delta^{n-1}/\partial\Delta^{n-1} \approx \partial\Delta^n/\Lambda$. The induction step then follows since the cycle i_n is sent under the first isomorphism to the cycle ∂i_n which equals $\pm i_{n-1}$ in $C_{n-1}(\partial\Delta^n, \Lambda)$.

To find a cycle generating $\widetilde{H}_n(S^n)$ let us regard S^n as two n-simplices Δ_1^n and Δ_2^n with their boundaries identified in the obvious way, preserving the ordering of vertices. The difference $\Delta_1^n - \Delta_2^n$, viewed as a singular n-chain, is then a cycle, and we claim it represents a generator of $\widetilde{H}_n(S^n)$, assuming $n > 0$ so that the latter group is infinite cyclic. To see this, consider the isomorphisms

$$\widetilde{H}_n(S^n) \xrightarrow{\approx} H_n(S^n, \Delta_2^n) \xleftarrow{\approx} H_n(\Delta_1^n, \partial\Delta_1^n)$$

where the first isomorphism comes from the long exact sequence of the pair (S^n, Δ_2^n) and the second isomorphism is justified by passing to quotients as before. Under these isomorphisms the cycle $\Delta_1^n - \Delta_2^n$ in the first group corresponds to the cycle Δ_1^n in the third group, which represents a generator of this group as we have seen, so $\Delta_1^n - \Delta_2^n$ represents a generator of $\widetilde{H}_n(S^n)$.

The preceding proposition implies that the excision property holds also for subcomplexes of CW complexes:

Corollary 2.24. *If the CW complex X is the union of subcomplexes A and B, then the inclusion $(B, A\cap B) \hookrightarrow (X, A)$ induces isomorphisms $H_n(B, A\cap B) \to H_n(X, A)$ for all n.*

Proof: Since CW pairs are good, Proposition 2.22 allows us to pass to the quotient spaces $B/(A\cap B)$ and X/A which are homeomorphic, assuming we are not in the trivial case $A \cap B = \varnothing$. ◻

Here is another application of the preceding proposition:

Corollary 2.25. *For a wedge sum $\bigvee_\alpha X_\alpha$, the inclusions $i_\alpha : X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce an isomorphism $\bigoplus_\alpha i_{\alpha *} : \bigoplus_\alpha \widetilde{H}_n(X_\alpha) \to \widetilde{H}_n(\bigvee_\alpha X_\alpha)$, provided that the wedge sum is formed at basepoints $x_\alpha \in X_\alpha$ such that the pairs (X_α, x_α) are good.*

Proof: Since reduced homology is the same as homology relative to a basepoint, this follows from the proposition by taking $(X, A) = (\coprod_\alpha X_\alpha, \coprod_\alpha \{x_\alpha\})$. ◻

Here is an application of the machinery we have developed, a classical result of Brouwer from around 1910 known as 'invariance of dimension,' which says in particular that $\mathbb{R}^m$ is not homeomorphic to $\mathbb{R}^n$ if $m \neq n$.

Theorem 2.26. *If nonempty open sets $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^n$ are homeomorphic, then $m = n$.*

Proof: For $x \in U$ we have $H_k(U, U - \{x\}) \approx H_k(\mathbb{R}^m, \mathbb{R}^m - \{x\})$ by excision. From the long exact sequence for the pair $(\mathbb{R}^m, \mathbb{R}^m - \{x\})$ we get $H_k(\mathbb{R}^m, \mathbb{R}^m - \{x\}) \approx \widetilde{H}_{k-1}(\mathbb{R}^m - \{x\})$. Since $\mathbb{R}^m - \{x\}$ deformation retracts onto a sphere S^{m-1}, we conclude that $H_k(U, U - \{x\})$ is $\mathbb{Z}$ for $k = m$ and 0 otherwise. By the same reasoning, $H_k(V, V - \{y\})$ is $\mathbb{Z}$ for $k = n$ and 0 otherwise. Since a homeomorphism $h : U \to V$ induces isomorphisms $H_k(U, U - \{x\}) \to H_k(V, V - \{h(x)\})$ for all k, we must have $m = n$. ◻

Generalizing the idea of this proof, the **local homology groups** of a space X at a point $x \in X$ are defined to be the groups $H_n(X, X - \{x\})$. For any open neighborhood U of x, excision gives isomorphisms $H_n(X, X - \{x\}) \approx H_n(U, U - \{x\})$, so these groups depend only on the local topology of X near x. A homeomorphism $f : X \to Y$ must induce isomorphisms $H_n(X, X - \{x\}) \approx H_n(Y, Y - \{f(x)\})$ for all x and n, so these local homology groups can be used to tell when spaces are not locally homeomorphic at certain points, as in the preceding proof. The exercises give some further examples of this.

Naturality

The exact sequences we have been constructing have an extra property that will become important later at key points in many arguments, though at first glance this property may seem just an idle technicality, not very interesting. We shall discuss the property now rather than interrupting later arguments to check it when it is needed, but the reader may prefer to postpone a careful reading of this discussion.

The property is called **naturality**. For example, to say that the long exact sequence of a pair is natural means that for a map $f:(X,A)\to(Y,B)$, the diagram

$$\begin{tikzcd}
\cdots \arrow[r] & H_n(A) \arrow[r, "i_*"] \arrow[d, "f_*"] & H_n(X) \arrow[r, "j_*"] \arrow[d, "f_*"] & H_n(X,A) \arrow[r, "\partial"] \arrow[d, "f_*"] & H_{n-1}(A) \arrow[r] \arrow[d, "f_*"] & \cdots \\
\cdots \arrow[r] & H_n(B) \arrow[r, "i_*"] & H_n(Y) \arrow[r, "j_*"] & H_n(Y,B) \arrow[r, "\partial"] & H_{n-1}(B) \arrow[r] & \cdots
\end{tikzcd}$$

is commutative. Commutativity of the squares involving i_* and j_* follows from the obvious commutativity of the corresponding squares of chain groups, with C_n in place of H_n. For the other square, when we defined induced homomorphisms we saw that $f_\sharp\partial = \partial f_\sharp$ at the chain level. Then for a class $[\alpha]\in H_n(X,A)$ represented by a relative cycle α, we have $f_*\partial[\alpha] = f_*[\partial\alpha] = [f_\sharp\partial\alpha] = [\partial f_\sharp\alpha] = \partial[f_\sharp\alpha] = \partial f_*[\alpha]$.

Alternatively, we could appeal to the general algebraic fact that the long exact sequence of homology groups associated to a short exact sequence of chain complexes is natural: For a commutative diagram of short exact sequences of chain complexes

$$\begin{tikzcd}
 & & 0 \arrow[dd] & & 0 \arrow[dd] & & 0 \arrow[dd] & \\
 & 0 \arrow[dd] & & 0 \arrow[dd] & & 0 \arrow[dd] & & \\
 & \cdots \arrow[r] & A'_{n+1} \arrow[rr, "\partial"] \arrow[dd, "i'"] & & A'_n \arrow[rr, "\partial"] \arrow[dd, "i'"] & & A'_{n-1} \arrow[r] \arrow[dd, "i'"] & \cdots \\
\cdots \arrow[r] & A_{n+1} \arrow[rr, "\partial"] \arrow[ur, "\alpha"] \arrow[dd, "i"] & & A_n \arrow[rr, "\partial"] \arrow[ur, "\alpha"] \arrow[dd, "i"] & & A_{n-1} \arrow[r] \arrow[ur, "\alpha"] \arrow[dd, "i"] & \cdots & \\
 & \cdots \arrow[r] & B'_{n+1} \arrow[rr, "\partial"] \arrow[dd, "j'"] & & B'_n \arrow[rr, "\partial"] \arrow[dd, "j'"] & & B'_{n-1} \arrow[r] \arrow[dd, "j'"] & \cdots \\
\cdots \arrow[r] & B_{n+1} \arrow[rr, "\partial"] \arrow[ur, "\beta"] \arrow[dd, "j"] & & B_n \arrow[rr, "\partial"] \arrow[ur, "\beta"] \arrow[dd, "j"] & & B_{n-1} \arrow[r] \arrow[ur, "\beta"] \arrow[dd, "j"] & \cdots & \\
 & \cdots \arrow[r] & C'_{n+1} \arrow[rr, "\partial"] \arrow[dd] & & C'_n \arrow[rr, "\partial"] \arrow[dd] & & C'_{n-1} \arrow[r] \arrow[dd] & \cdots \\
\cdots \arrow[r] & C_{n+1} \arrow[rr, "\partial"] \arrow[ur, "\gamma"] \arrow[dd] & & C_n \arrow[rr, "\partial"] \arrow[ur, "\gamma"] \arrow[dd] & & C_{n-1} \arrow[r] \arrow[ur, "\gamma"] \arrow[dd] & \cdots & \\
 & & 0 & & 0 & & 0 & \\
 & 0 & & 0 & & 0 & &
\end{tikzcd}$$

the induced diagram

$$\begin{tikzcd}
\cdots \arrow[r] & H_n(A) \arrow[r, "i_*"] \arrow[d, "\alpha_*"] & H_n(B) \arrow[r, "j_*"] \arrow[d, "\beta_*"] & H_n(C) \arrow[r, "\partial"] \arrow[d, "\gamma_*"] & H_{n-1}(A) \arrow[r] \arrow[d, "\alpha_*"] & \cdots \\
\cdots \arrow[r] & H_n(A') \arrow[r, "i'_*"] & H_n(B') \arrow[r, "j'_*"] & H_n(C') \arrow[r, "\partial"] & H_{n-1}(A') \arrow[r] & \cdots
\end{tikzcd}$$

is commutative. Commutativity of the first two squares is obvious since $\beta i = i'\alpha$ implies $\beta_* i_* = i'_*\alpha_*$ and $\gamma j = j'\beta$ implies $\gamma_* j_* = j'_*\beta_*$. For the third square, recall that the map $\partial: H_n(C)\to H_{n-1}(A)$ was defined by $\partial[c] = [a]$ where $c = j(b)$ and $i(a) = \partial b$. Then $\partial[\gamma(c)] = [\alpha(a)]$ since $\gamma(c) = \gamma j(b) = j'(\beta(b))$ and $i'(\alpha(a)) = \beta i(a) = \beta\partial(b) = \partial\beta(b)$. Hence $\partial\gamma_*[c] = \alpha_*[a] = \alpha_*\partial[c]$.

This algebraic fact also implies naturality of the long exact sequence of a triple and the long exact sequence of reduced homology of a pair.

Finally, there is the naturality of the long exact sequence in Theorem 2.13, that is, commutativity of the diagram

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & \widetilde{H}_n(A) & \xrightarrow{i_*} & \widetilde{H}_n(X) & \xrightarrow{q_*} & \widetilde{H}_n(X/A) & \xrightarrow{\partial} & \widetilde{H}_{n-1}(A) & \longrightarrow & \cdots \\ & & \downarrow f_* & & \downarrow f_* & & \downarrow \overline{f}_* & & \downarrow f_* & & \\ \cdots & \longrightarrow & \widetilde{H}_n(B) & \xrightarrow{i_*} & \widetilde{H}_n(Y) & \xrightarrow{q_*} & \widetilde{H}_n(Y/B) & \xrightarrow{\partial} & \widetilde{H}_{n-1}(B) & \longrightarrow & \cdots \end{array}$$

where i and q denote inclusions and quotient maps, and $\overline{f}: X/A \to Y/B$ is induced by f. The first two squares commute since $fi = if$ and $\overline{f}q = qf$. The third square expands into

$$\begin{array}{ccccccc} \widetilde{H}_n(X/A) & \underset{\approx}{\xrightarrow{j_*}} & H_n(X/A, A/A) & \underset{\approx}{\xleftarrow{q_*}} & H_n(X,A) & \xrightarrow{\partial} & \widetilde{H}_{n-1}(A) \\ \downarrow \overline{f}_* & & \downarrow \overline{f}_* & & \downarrow f_* & & \downarrow f_* \\ \widetilde{H}_n(Y/B) & \underset{\approx}{\xrightarrow{j_*}} & H_n(Y/B, B/B) & \underset{\approx}{\xleftarrow{q_*}} & H_n(Y,B) & \xrightarrow{\partial} & \widetilde{H}_{n-1}(B) \end{array}$$

We have already shown commutativity of the first and third squares, and the second square commutes since $\overline{f}q = qf$.

The Equivalence of Simplicial and Singular Homology

We can use the preceding results to show that the simplicial and singular homology groups of Δ-complexes are always isomorphic. For the proof it will be convenient to consider the relative case as well, so let X be a Δ-complex with $A \subset X$ a subcomplex. Thus A is the Δ-complex formed by any union of simplices of X. Relative groups $H_n^\Delta(X,A)$ can be defined in the same way as for singular homology, via relative chains $\Delta_n(X,A) = \Delta_n(X)/\Delta_n(A)$, and this yields a long exact sequence of simplicial homology groups for the pair (X,A) by the same algebraic argument as for singular homology. There is a canonical homomorphism $H_n^\Delta(X,A) \to H_n(X,A)$ induced by the chain map $\Delta_n(X,A) \to C_n(X,A)$ sending each n-simplex of X to its characteristic map $\sigma: \Delta^n \to X$. The possibility $A = \varnothing$ is not excluded, in which case the relative groups reduce to absolute groups.

Theorem 2.27. *The homomorphisms $H_n^\Delta(X,A) \to H_n(X,A)$ are isomorphisms for all n and all Δ-complex pairs (X,A).*

Proof: First we do the case that X is finite-dimensional and A is empty. For X^k the k-skeleton of X, consisting of all simplices of dimension k or less, we have a commutative diagram of exact sequences:

$$\begin{array}{ccccccccc} H_{n+1}^\Delta(X^k,X^{k-1}) & \longrightarrow & H_n^\Delta(X^{k-1}) & \longrightarrow & H_n^\Delta(X^k) & \longrightarrow & H_n^\Delta(X^k,X^{k-1}) & \longrightarrow & H_{n-1}^\Delta(X^{k-1}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H_{n+1}(X^k,X^{k-1}) & \longrightarrow & H_n(X^{k-1}) & \longrightarrow & H_n(X^k) & \longrightarrow & H_n(X^k,X^{k-1}) & \longrightarrow & H_{n-1}(X^{k-1}) \end{array}$$